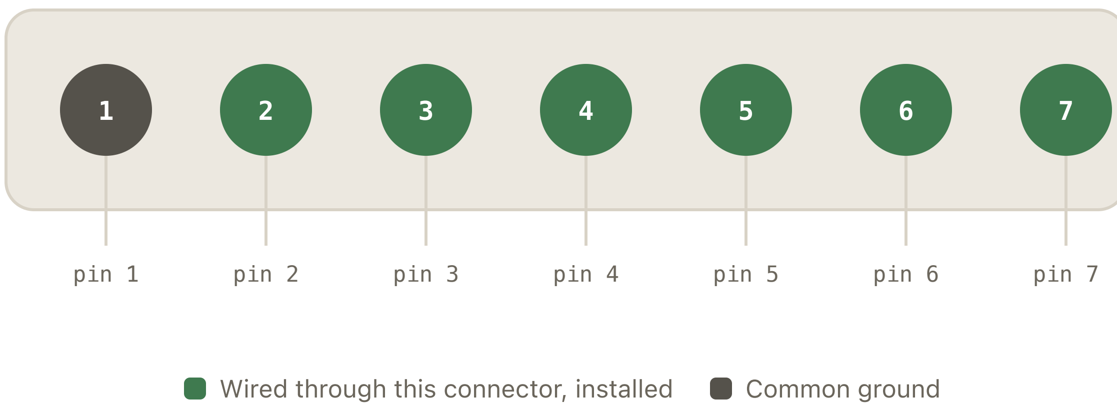
LR2-REDUX · HARDWARE REFERENCE

7-pin harness connector

Confirmed by direct multimeter continuity trace (2026-07-03) and bench testing (2026-07-04/05); hall sensors finalized in the chassis (2026-07-06). All 7 pins are wired through the stock harness connector as installed in the actual LR2 — see `CLAUDE.md` for the full trace/decision history.



PIN	FUNCTION	ESP32 PIN	STATUS
1	GND common return — both hall sensors and both switches	**GND**	Ground
2	Hall “B” OUT via 330Ω pull-up network — Dump position	**GPIO35**	Wired
3	Hall “A” OUT via 330Ω pull-up network — Home position	**GPIO34**	Wired
4	Hall “A” VCC shared PIC-supplied line	**5V**	Wired
5	Hall “B” VCC same net as pin 4 — bridged trace	**5V**	Wired
6	Weight switch PIN_WEIGHT_SWITCH — INPUT_PULLUP, active-low	**GPIO32**	Wired

PIN	FUNCTION	ESP32 PIN	STATUS
7	Anti-pinch switch PIN_ANTI_PINCH — INPUT_PULLUP, active-HIGH	**GPIO14**	Wired

> **All 7 pins are now finalized and installed.** The hall sensors are mounted in their permanent positions in the actual LR2 chassis and wired through this stock connector, not on the breadboard. The original board's pin 2/3 labels ("A"/"B") turned out to map onto Home/Dump as: pin 3 ("A" OUT) → GPIO34 (Home), pin 2 ("B" OUT) → GPIO35 (Dump). Pin 7 (anti-pinch) moved from GPIO13 to **GPIO14** (2026-07-06) — the ESP32 mini board in use now doesn't break out GPIO13 on its header.

GPIO34/35 are input-only pins with no internal pull capability — both hall sensors need external 10kΩ pull-ups to 3.3V (not 5V — the ESP32 isn't 5V-tolerant, but the A1101EUA-T hall ICs need ≥4.5V to run, so the sensor is powered from 5V and only its open-drain output is pulled up to 3.3V). GPIO32/14 use the ESP32's internal `INPUT_PULLUP`. Pins 6/7 were split from a shared series loop on the original board — see the anti-pinch polarity note in `CLAUDE.md` (it's active-HIGH, opposite of every other input here).